LeafLoom

Grow with confidence.

AI Plant Care & Smart Gardening

Product, design and technical documentation — every feature, screen and decision in one place.



COURSE	SE4041 Mobile Application Design & Development — Assignment 02
PLATFORMS	iOS 17 (SwiftUI) · visionOS 2 (Part B) · Windows browser preview
EDITION	9 October 2026

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ABOUT THE SCREENSHOTS

All screenshots are real captures of the LeafLoom browser preview, which mirrors the native SwiftUI screens, data model and rules. In the preview, plant identification and reminders are simulated and clearly labelled. The native iOS and visionOS code has been written but not yet compiled — that needs Xcode on a Mac.

01

Overview

LeafLoom is a calm, private iPhone companion for houseplant owners. It brings plant records, identification help, scheduled care and a growth journal together — on device, with no account.

THE PROBLEM

Plant care is scattered

Owners struggle to identify plants, understand each one's light, water, humidity and soil needs, remember recurring care, and notice change over time. The information lives across labels, search engines, notes and calendar reminders.

THE SOLUTION

One local-first home for every plant

A structured collection with photos, care tasks that repeat sensibly and remind you, on-device identification with Core ML, a growth journal with before/after photos, and insights computed from your own records.

Objectives

#	Objective	Measured by
O1	Keep a structured, searchable plant collection	Plant CRUD, search, sort, filter and favourites all persist
O2	Assist identification with on-device ML	Core ML/Vision behind a protocol; clearly labelled demo mode without a model
O3	Make recurring care easy to remember	Care tasks with recurrence, completion history and local notifications
O4	Make change visible over time	Growth journal with photos, heights and a timeline
O5	Show collection health at a glance	Dashboard and Insights computed from stored records
O6	Be accessible and pleasant	Dynamic Type, Dark Mode, VoiceOver labels, Reduce Motion

Who it is for

Segment	Need	How LeafLoom helps
Indoor plant owners	Remember watering; keep plants alive	Recurring reminders and a "needs attention" dashboard
Beginner gardeners	Learn what a plant is and needs	Identification with care information; guided add-plant form

Segment	Need	How LeafLoom helps
Students into sustainability	Low-cost, approachable start	Free, offline, no account
Enthusiasts with big collections	Organise and monitor many plants	Search, filters, tags, locations and insights

Deliverables

Deliverable	Folder	Status
Native SwiftUI app (iOS 17)	`native-ios/`	Source complete · needs Mac to compile
Windows browser preview (React + TypeScript)	`windows-preview/`	Verified on Windows
Part B — LeafLoom Spatial (visionOS 2)	`partB-visionos/`	Source & docs complete · needs Mac
Documentation set	`docs/`	Complete

02

Design language

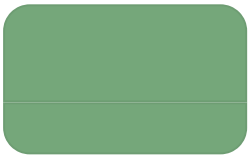
Calm and botanical: a deep forest green with sage, amber and terracotta accents, rounded cards on warm cream surfaces, editorial serif headlines and original illustrations.

Colour — light and dark

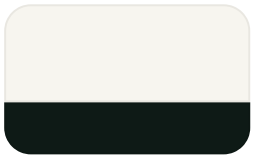
Every colour is a semantic token with a designed dark variant (deep botanical surfaces, not an inversion). The browser preview's tokens are the source of truth; a script generates the 26 colour sets in the iOS asset catalog from them.



Primary
#214e3d · #8fc495



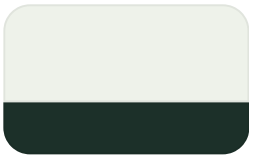
Accent
#75a77a · #75a77a



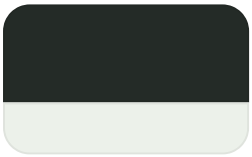
Background
#f7f5ef · #0e1a16



Surface
#ffffff · #15251f



Surface alt
#eef2ea · #1c3029



Text
#242b27 · #ecf1ea



Amber
#ddaa52 · #e2b566



Danger
#b3261e · #f2726b



Success
#24653d · #7ccb91



Info
#2f5f8a · #8db8e0

Typography

Good morning

Large titles — New York serif (Georgia fallback)

This week's care

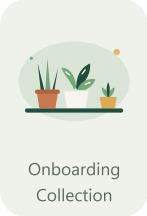
Headlines and body — SF Pro (system) at Dynamic Type sizes: 34 / 28 / 22 / 20 / 17 / 15 / 13 / 12 pt

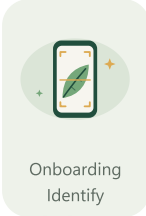
Shape & spacing

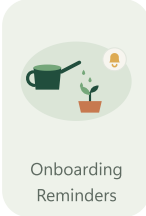
- 4-pt spacing scale: 4, 8, 12, 16, 24, 32, 48
- Radii 8 / 12 / 18 / 26 pt and pill
- 44 × 44 pt minimum touch targets
- Soft, low-contrast card shadows
- Motion 150–420 ms, removed under Reduce Motion

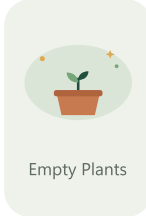
Original illustrations

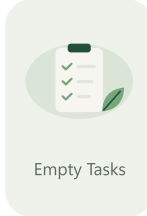
Drawn for LeafLoom as SVG and shipped as vector assets — used for onboarding and every empty state.

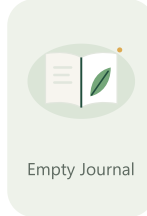

Onboarding Collection

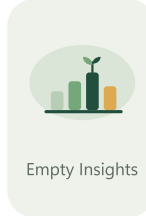

Onboarding Identify

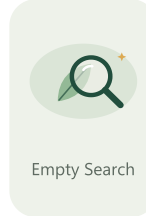

Onboarding Reminders


Empty Plants


Empty Tasks


Empty Journal


Empty Insights


Empty Search


App icon — rasterised to 1024 px for the asset catalog

Sample photography

Eight AI-generated plant photographs illustrate the sample collection. They are listed with their origin in `ASSET_ATTRIBUTION.md`.



03

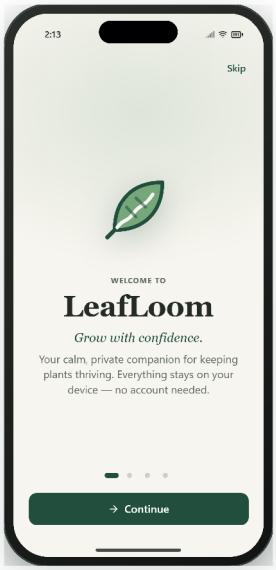
Feature tour

A walk through every screen of LeafLoom, from first launch to settings. Five tabs — Home, My Plants, Identify, Care and Insights — each keep their own navigation stack; re-tapping a tab returns to its first screen.

FIRST LAUNCH

Onboarding

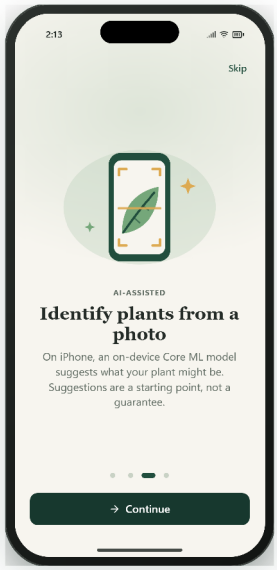
Four short pages welcome the user and explain the three pillars of the app. Notification permission is only requested here, from the reminders page, after the benefit is explained.



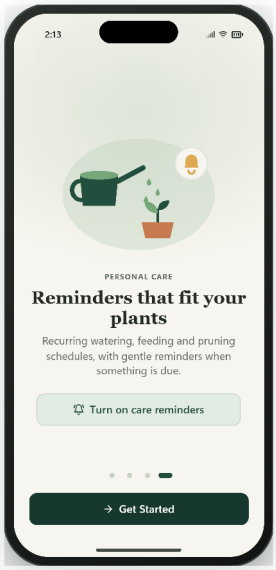
Welcome



Your collection



Identify from a photo



Reminders

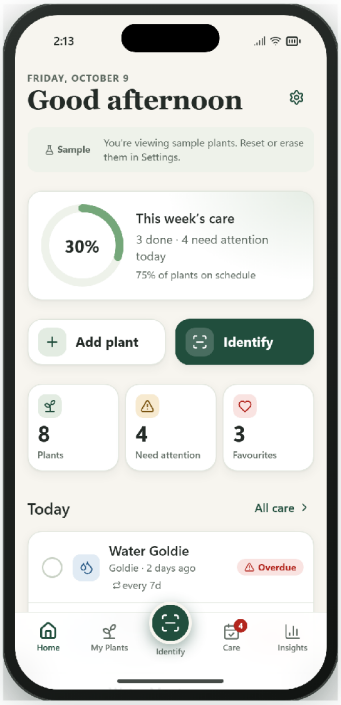
- Skip at any time; progress dots show where you are
- The identify page states plainly that a demo mode is used without a Core ML model
- Completion is remembered; replay from Settings
- Starts with eight labelled sample plants so there is something to explore

HOME TAB

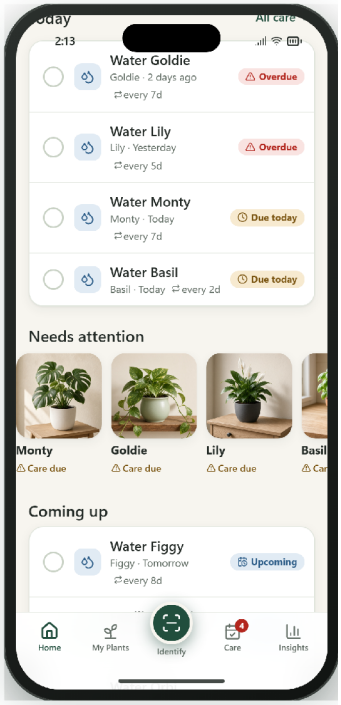
Dashboard

A personal, glanceable summary. Every number is calculated from stored records — nothing is decorative.

- Greeting with today's date and a sample-data banner
- Weekly care ring: completed ÷ (completed + open tasks due this week, including overdue); weeks start on Monday
- Quick actions: Add plant and Identify
- Statistic cards: plants, needing attention, favourites
- **Today** list with one-tap completion and **Undo**
- Needs attention, Coming up (next 7 days) and Recently added
- Gear button opens Settings



Greeting, weekly ring, quick actions



Today, attention, coming up

MY PLANTS TAB

Your collection

A searchable, filterable collection shown as a photo grid or a compact list.



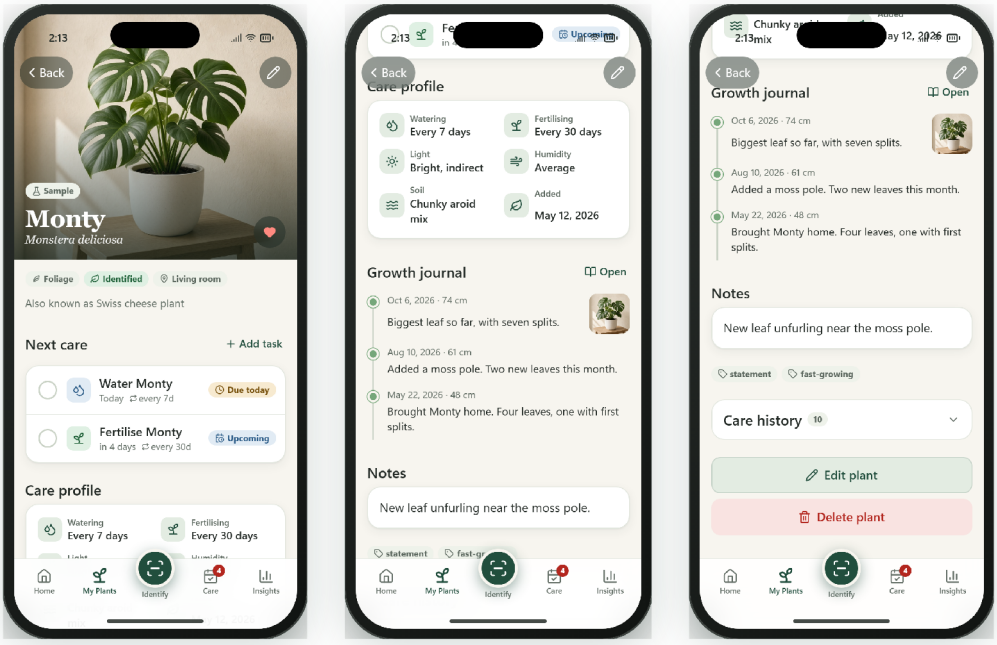
Grid layout | Filters panel | List layout

- Search by name, common name, scientific name or tag (case- and accent-insensitive)
- Sort: Recently added, A–Z, Next care
- Filter by category, location and favourites; Clear filters
- Heart a plant to favourite it from the card
- Grid/list toggle, also a setting
- Designed empty and no-results states

PLANT DETAILS

Everything about one plant

A hero photo with the plant's identity, followed by care, journal and history.



Hero, status and next care | Care profile and journal | Notes, tags and history

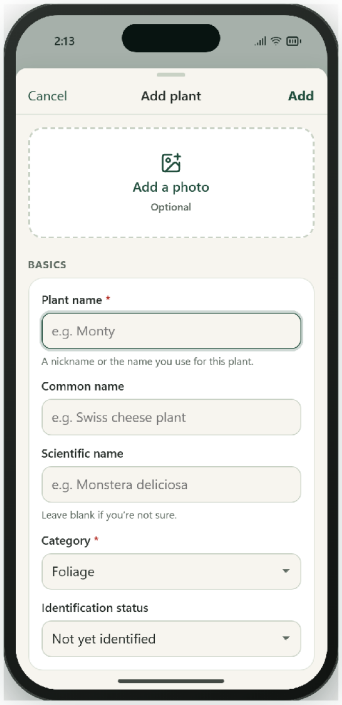
- Sample badge and favourite button on the hero
- Next care with **Add task**
- Care profile: watering, fertilising, light, humidity, soil, date added
- Growth journal preview with **Open**
- Notes, tags and collapsible care history
- **Edit** and **Delete plant** — deletion confirms and states how many tasks and entries go with it

CREATE & EDIT

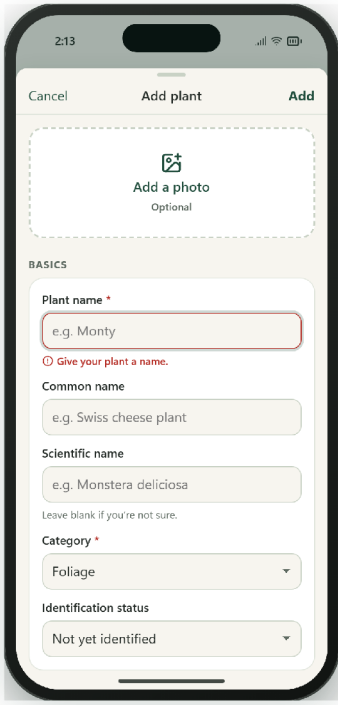
Adding a plant, with validation

One form for adding and editing. Mistakes are explained under the field they belong to.

- Required: name (1–60 characters after trimming) and category
- Optional: names, photo (library or camera), location, acquisition date (not in the future), light, humidity, soil, notes, tags
- Watering/fertilising intervals of 1–365 days create repeating care tasks automatically
- Tags are normalised: lower-cased, de-duplicated and capped
- Closing with unsaved changes asks before discarding



Add plant form

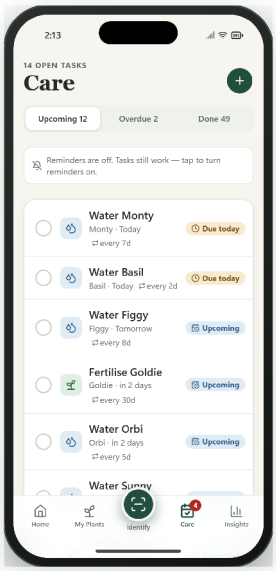


Inline validation

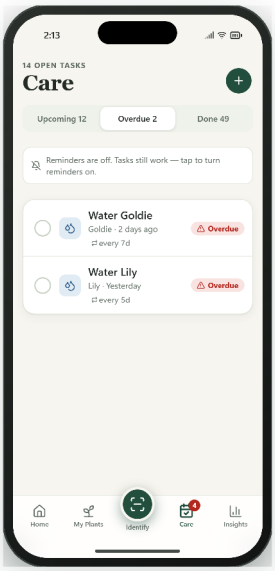
CARE TAB

Care tasks that repeat sensibly

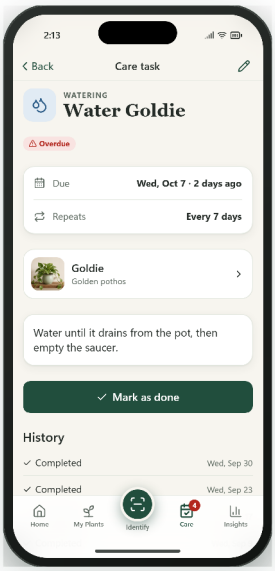
Watering, fertilising, pruning, repotting, cleaning, inspection or custom tasks — grouped by day, with overdue work surfaced first.



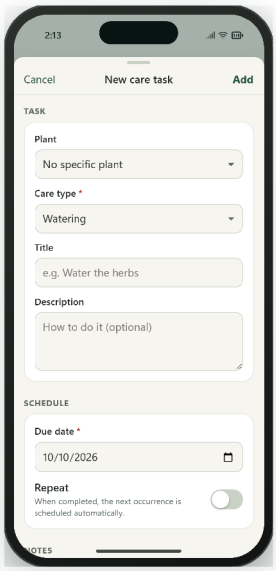
Upcoming



Overdue



Task details



New task

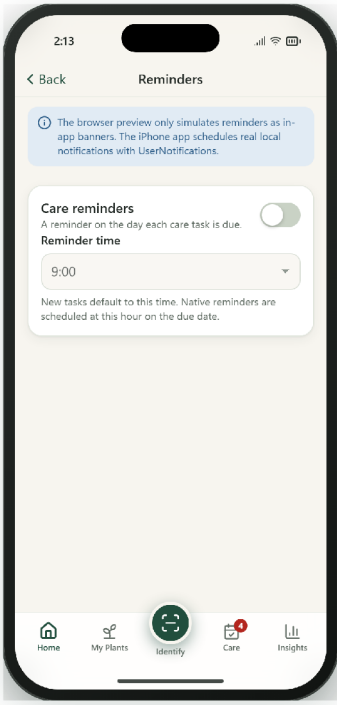
- Segments for upcoming, overdue and completed, with live counts
- Complete, undo, edit and delete (with confirmation); swipe actions natively
- **Recurrence:** the next occurrence is due at the start of the completion day + interval, at the original time of day — late care never creates a backlog
- **Undo** removes the spawned occurrence only if it is still untouched
- Task details show the whole repeating series
- Suggested titles follow the category until you type your own

NOTIFICATIONS

Reminders

Local notifications at a time the user chooses. LeafLoom works fully without them.

- One reminder per open task on its due date, at the preferred hour (6:00–22:00)
- Edits reschedule; completion and deletion cancel
- Clear permission states, with a shortcut to system Settings when denied
- In the browser preview, reminders are simulated in-app and labelled as such

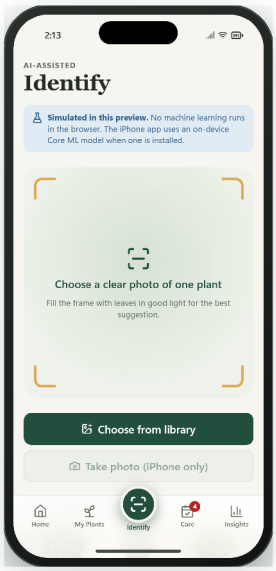


Reminder settings

IDENTIFY TAB · CORE ML

Identify a plant from a photo

On device, with Core ML and Vision. Without a bundled model a clearly labelled demo mode shows how it works — and never pretends to be confident.



Choose a photo

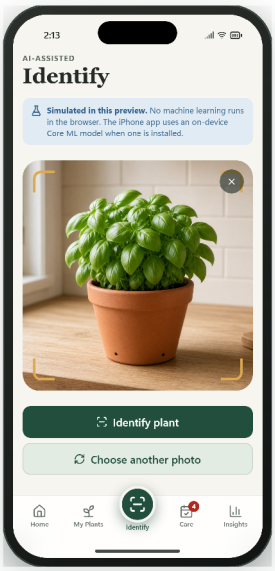
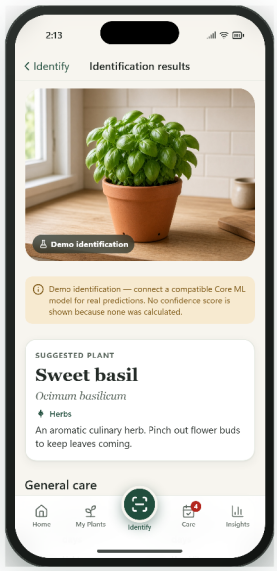
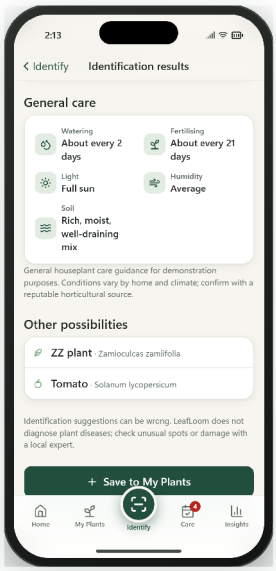


Photo selected



Labelled demo result



Care, alternatives, save

- Photo library (no permission needed), camera on real devices, or bundled samples
- Analysis runs off the main thread on a 1024-px downsampled image
- Top suggestion, general care guidance and up to four alternatives
- Confidence appears only from a real model; below 40% a low-confidence warning is shown
- **Save to My Plants** opens a prefilled form for review; identification history is kept
- No disease diagnosis claims

HONEST AI

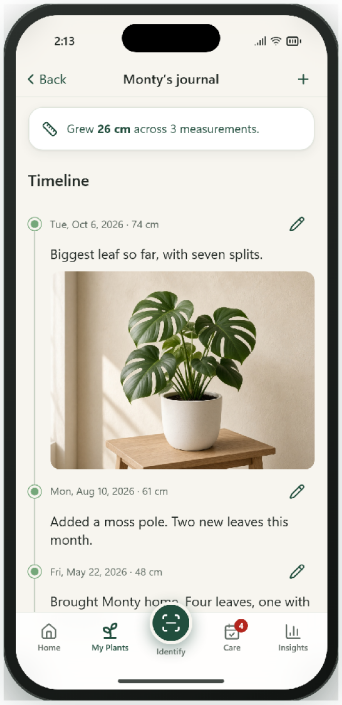
No plant model is included and no accuracy is claimed. Demo results carry a "Demo identification" banner and never show a percentage. `CORE_ML_GUIDE.md` explains how to train `LeafLoomPlantClassifier` with Create ML and drop it in.

GROWTH JOURNAL

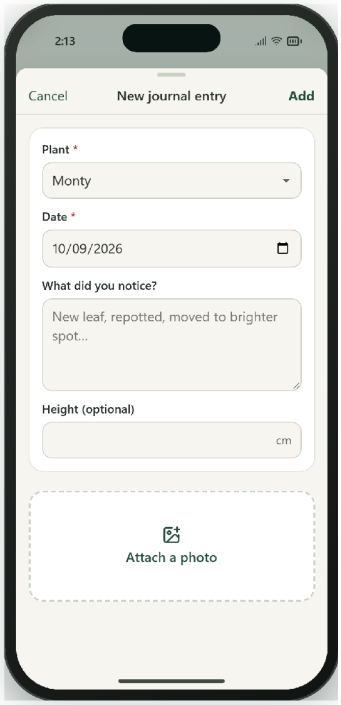
See growth over time

Dated notes with optional photos and heights, shown as a timeline.

- Notes required; photo and height (cm) optional; dates cannot be in the future
- Before/after slider compares the earliest and latest photos (adjustable with VoiceOver)
- Growth summary such as "Grew 26 cm across 3 measurements"
- Edit and delete entries, with confirmation



Timeline and before/after

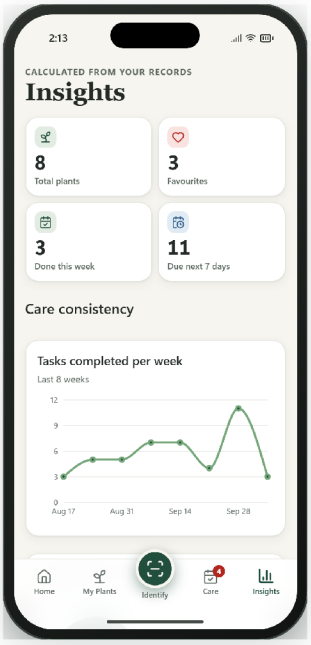


New entry

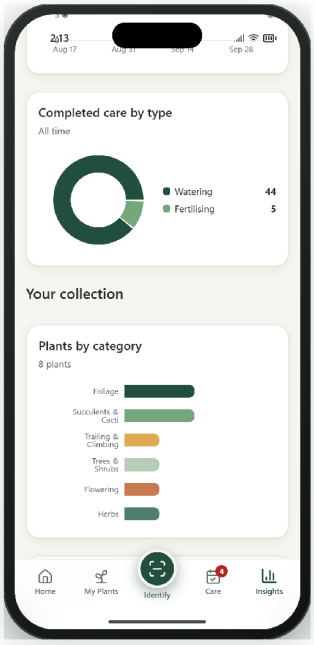
INSIGHTS TAB

Insights

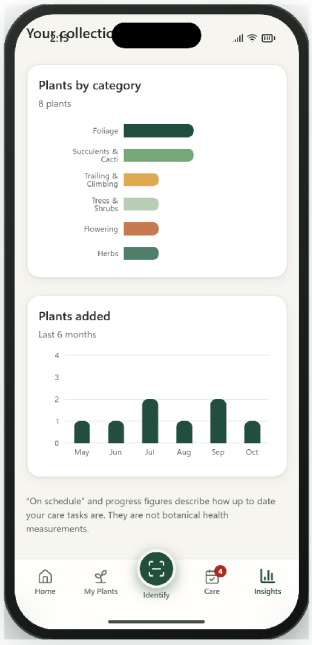
Charts built from the user's own records, each with a spoken summary for VoiceOver.



Stats and completion trend



Care mix



Categories and additions

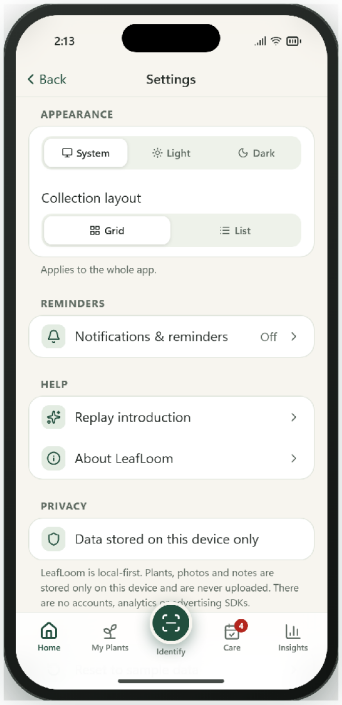
- Statistic cards for the collection and this week
- Care completion trend over the last eight Monday-start weeks
- Mix of completed care by type
- Plants by category and plants added per month (last six months)
- Charts stop animating under Reduce Motion
- Copy is careful: "on schedule" describes task status, not botanical health

SETTINGS

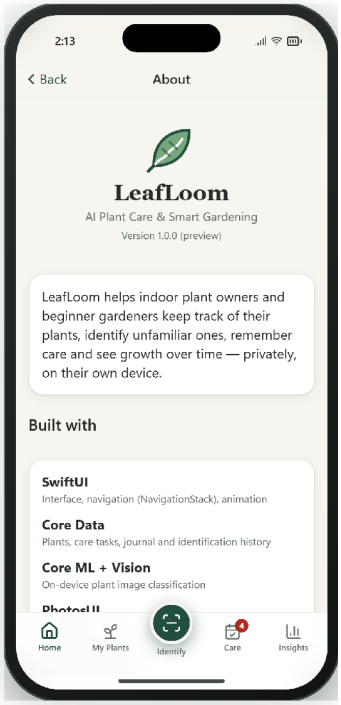
Settings & About

Appearance, reminders, help, privacy and data controls in one place.

- Theme: System, Light or Dark — applied app-wide
- Default collection layout: grid or list
- Notifications & reminders
- Replay the introduction
- **Reset to sample data** and **Erase all data**, each behind a confirmation
- About: privacy, how identification works, care disclaimer, credits, version and identification mode



Settings

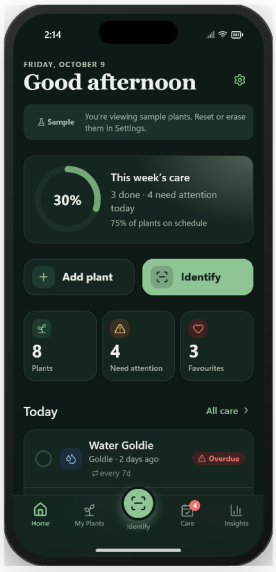


About LeafLoom

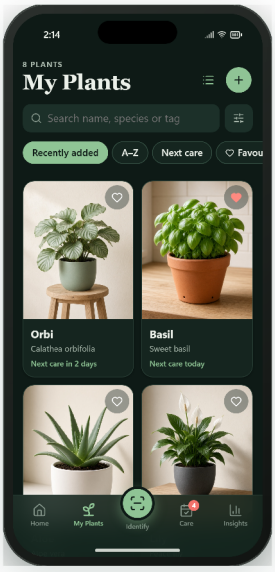
APPEARANCE

Dark Mode

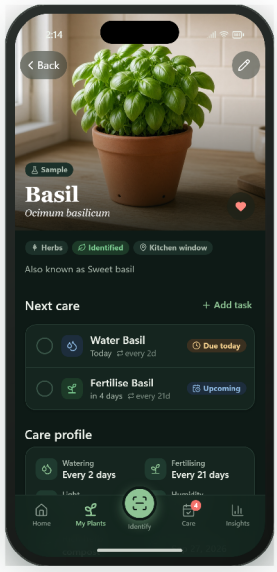
A designed dark theme with deep botanical surfaces and lifted greens — every colour token has a dark counterpart.



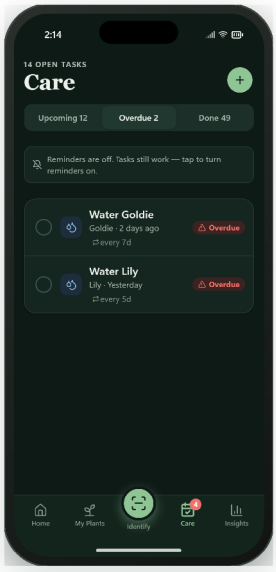
Home



My Plants



Plant details



Care

- Follows the system or can be forced in Settings
- Text keeps WCAG AA contrast on dark surfaces
- Photos sit on dark cards without harsh borders

04

Accessibility & privacy

Accessibility and privacy are defaults, not settings.

Accessibility

- **VoiceOver:** meaningful labels and hints on every control; charts have spoken summaries; the before/after slider is adjustable
- **Dynamic Type:** text styles throughout, layouts reflow up to the largest accessibility sizes
- **Not colour alone:** overdue, due and fit states always pair text with an icon
- **Reduce Motion:** decorative animation and chart animation are removed
- **Targets:** at least 44 × 44 pt
- **Dark Mode:** designed, not inverted

Privacy by design

- Plants, photos and journal entries are stored only on the device
- No account, no analytics, no server, no network calls
- Identification runs entirely on device
- Photos are downsampled JPEG files in Application Support
- Notification permission is asked in context, and the app works fully without it

CARE DISCLAIMER

Care tips are general guidance. Identification suggestions can be wrong — never use them to decide whether a plant is safe to eat or safe for pets.

05

Architecture

A layered SwiftUI app: views read reactively, repositories own every write, domain rules are pure Swift, and framework services sit behind protocols.

Technology

Concern	Choice	Notes
UI	SwiftUI, iOS 17	`NavigationStack`, `@Observable`, `ContentUnavailableView`, sensory feedback
Persistence	Core Data	Four entities, typed accessors over `…Raw` attributes, lightweight migration
Machine learning	Core ML + Vision	`VNCoreMLRequest`, centre crop, top-5 results off the main actor
Photos	PhotosUI `PhotosPicker`	Out-of-process picker, no library permission
Camera	`UIImagePickerController`	Only UIKit interop; shown when a camera exists
Notifications	UserNotifications	Calendar triggers; identifier = task UUID
Charts	Swift Charts	Line, area, bar and sector marks
Tests	XCTest / XCUITest	In-memory store; launch arguments for deterministic UI tests

Layers

Presentation — Feature views · Components & modifiers · @FetchRequest reads

State — AppRouter (tab + stacks + sheet) · Form & flow view models · AppPreferences

Domain — pure Swift — PlantValidator · RecurrenceCalculator · PlantQuery · InsightsCalculator

Services — @MainActor repositories · ImageStore · PlantIdentifying · ReminderScheduling

Apple frameworks — Core Data · Core ML + Vision · UserNotifications · File system

Views never call `save()`. `AppEnvironment` is the dependency container; `router`, `toasts` and `preferences` are injected separately.

Core Data model

Plant
id · name · commonName · scientificName
categoryRaw · location · tagsRaw · notes
imageFilename · sampleAssetName
wateringIntervalRaw · fertilizingIntervalRaw
sunlightRaw · humidityRaw · soilPreference
isFavorite · isSample · identificationStatusRaw
dateAdded · acquisitionDate · createdAt · updatedAt

CareTask
id · seriesID · spawnedTaskID
title · descriptionText · categoryRaw
dueDate · completedAt · isCompleted
recurrenceIntervalRaw · notes · isSample
createdAt · updatedAt
Plant → tasks: cascade

JournalEntry
id · entryDate · notes
imageFilename · heightRaw
createdAt · updatedAt
Plant → entries: cascade

IdentificationRecord
id · dateCreated · imageFilename
predictedLabel · confidenceRaw
modelIdentifier · isDemoResult
Plant → records: nullify

Interval 0 means "not set"; height and confidence −1 mean "not supplied" — the Swift accessors expose them as optionals. Images live on disk, not in the store. Deleting a plant also cancels its notifications and removes image files.

Identification pipeline

1. Pick a photo (PhotosPicker, camera or sample)
2. Validate and downsample to 1024 px
3. `PlantIdentifying.identify` — Core ML if `LeafLoomPlantClassifier.mlmodelc` is bundled, otherwise demo
4. `LabelMapper` maps labels to the 14-entry knowledge base
5. Result shown; history saved; optional prefilled plant form

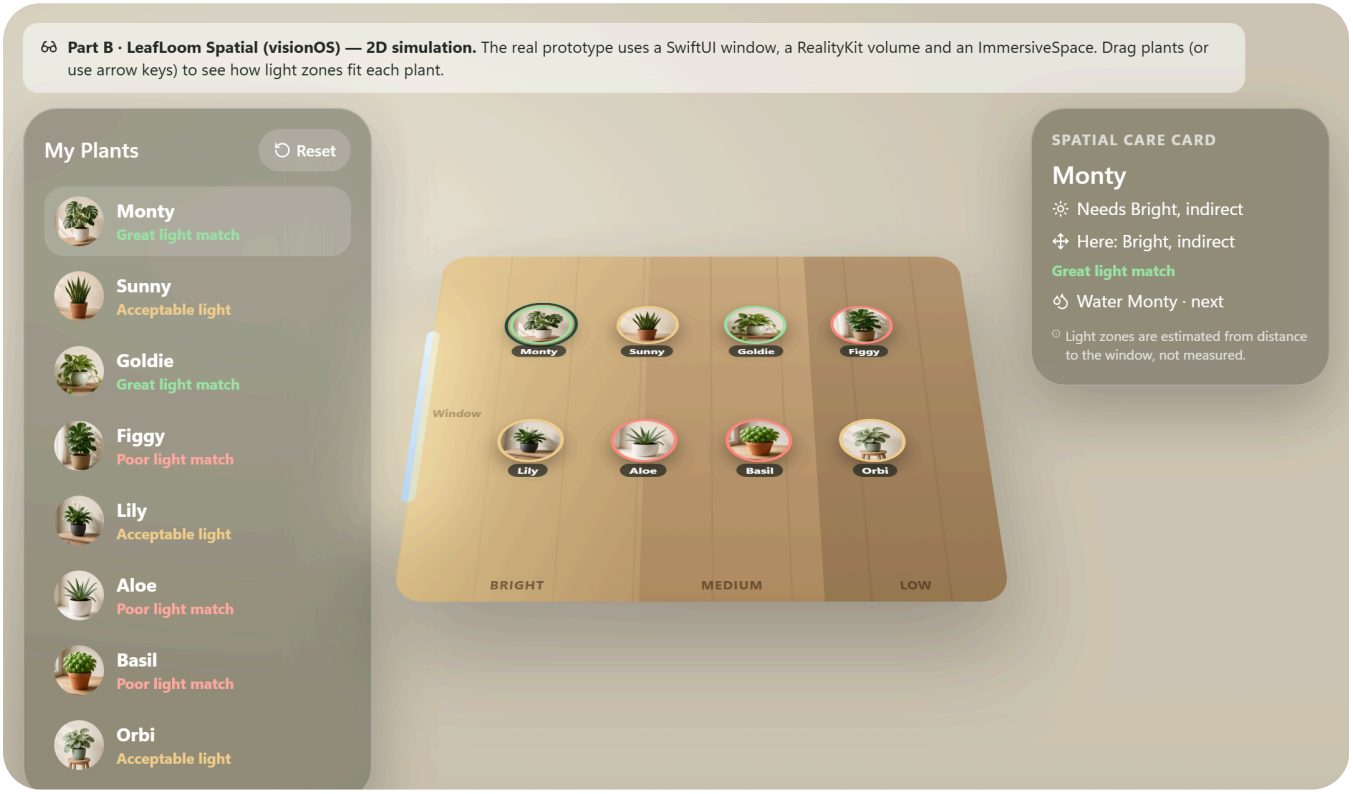
Navigation

- `AppRouter` holds the selected tab, one `NavigationPath` per tab and one optional sheet
- Push routes: plant details, task details, journal, settings, reminders, about
- Sheets: add/edit plant, task and journal entry — the sheet host passes the saved record in
- Saving from Identify switches to My Plants and opens the new plant

06

Part B — LeafLoom Spatial

A visionOS 2 prototype that answers the question phone apps cannot: will this plant thrive *there*? Arrange your own plants life-size in your room and see each spot's light fit before you move a pot.



The browser preview's 2D simulation of the concept: window, room floor plan with light zones, and a spatial care card.

Scene	visionOS type	What it does
Garden window	`WindowGroup`	Plant list with light-fit badges, care card with advice and next watering, "View in 3D", "Watered today"; an ornament holds Arrange in my room and Reset layout
Planter volume	Volumetric `WindowGroup`	One plant on a turntable — drag to spin, pinch to zoom, adjustable with VoiceOver
Room	`ImmersiveSpace` (mixed)	Plants on the real floor: drag, rotate, scale (0.5–2×), tap to select; a draggable window marker and floor rings show light zones

Light estimate

Distance from window	Zone
< 0.6 m	Full sun
0.6 – 1.5 m	Bright indirect
1.5 – 3.0 m	Medium
≥ 3.0 m	Low

Same zone = great match, one step = acceptable, further = poor — with "move closer / further" advice. It is a distance heuristic, labelled as an estimate.

Built with

- Procedural plants from RealityKit primitives — no third-party 3D assets
- Shapes seeded from each plant's UUID, stable every launch
- Targeted drag, rotate, magnify and tap gestures; layout saved once per gesture
- Billboarded attachments; accessibility components on entities
- Same eight sample plants as the iPhone app; 12 unit tests

Similar apps

App	Known for	Gap LeafLoom Spatial targets
PictureThis	Photo identification, care guides, reminders	No in-room placement
Planta	Adaptive schedules; camera light meter	Light is read one spot at a time on a phone
Pl@ntNet · iNaturalist/Seek	Free, research-backed identification	Identification only
IKEA / Houzz AR	True-scale placement in your room	Fit, not plant health

Written from general product knowledge — check current App Store listings before submission (see `COMPETITORS.md`).

Pitch

VALUE

See where every plant will thrive before you move a single pot. Fewer losses, less guesswork, one collection everywhere.

REVENUE (PROPOSAL)

Free up to 5 plants · **Plus** unlimited plants, rooms, sync · **Pro** for designers and shops. No ads, no data selling.

GO-TO-MARKET

Prototype → TestFlight beta with plant and Vision Pro communities → App Store launch with the iPhone app → plant-shop pilots.

07

Testing & quality

Only commands that were actually run are reported as results. Native suites are written and waiting for a Mac.

Verified on Windows — final code

Check	Result
ESLint	Pass no problems
Vitest unit & component tests	Pass 58 / 58 across 8 files
Type-check + production build	Pass index 364.6 kB (112.2 kB gzip), charts 413.2 kB (119.5 kB gzip)
Playwright end-to-end flows (Chromium)	Pass 5 / 5
Screenshot capture	Pass preview and phone captures, inspected

Defects found by testing — all fixed

Found by	Defect	Fix
Playwright	Crash after onboarding: an effect returned the Promise from `scrollTo` (jsdom hid it)	Block-bodied effects
Screenshots	Filter panel always visible — CSS overrode `hidden`	Global `[hidden]` rule
Playwright	Only 6 of 8 sample photos on Identify	Show every sample
Screenshots	Charts captured mid-animation, ignoring Reduce Motion	Disable chart animation
Build	Empty manual chunk	Removed
Asset script	Lost named export from an ESM import	`createRequire`

Native suites — written, not yet run

Suite	Tests	Covers	Status
LeafLoomTests	84	Validation, recurrence and status, queries, insights, identification and label mapping, reminders, images, all repositories, maintenance, form view models	Needs Mac
LeafLoomUITests	7	Onboarding skip, tabs, add plant with validation, undo, labelled demo identification, empty state, delete confirmation	Needs Mac
LeafLoomSpatialTests	12	Light zones and fit, sample data, procedural seed, store layout and persistence	Needs Mac

User acceptance tests (eight scripted scenarios) and a manual accessibility checklist are in `TEST_PLAN.md`, ready for real participants.

08

Running the project

Try LeafLoom on Windows today; build the native apps on a Mac with one script.

Browser preview (any OS)

```
cd windows-preview
npm install
npm run dev        # http://localhost:5173
npm run lint
npm test
npm run build
npm run e2e
```

The tools panel switches device size, theme, text size and Reduce Motion, jumps between screens, resets data and opens the Spatial simulation.

On a Mac (Xcode 16+)

1. Copy the folder via OneDrive, zip or USB — no git needed (`WINDOWS_TO_MAC_GUIDE.md`)
2. `chmod +x scripts/verify-native.sh`
3. `./scripts/verify-native.sh` — builds and tests iOS, then visionOS
4. Open `native-ios/LeafLoom.xcodeproj` and press ⌘R
5. Record real results in `TEST_PLAN.md` and the checklist

Launch arguments: `-skipOnboarding`, `-resetData`, `-emptyState`, `-uiTesting`.

Project map

Folder	Contents
`native-ios/`	LeafLoom.xcodeproj (synchronized folders), project.yml; App, Core, Data, Features, Services, Resources; unit and UI tests
`windows-preview/`	React 19 + TypeScript + Vite; IndexedDB repository; Vitest and Playwright
`partB-visionos/`	LeafLoomSpatial.xcodeproj, project.yml, sources, tests, CONCEPT / COMPETITORS / PITCH
`docs/`	Requirements, architecture, design system, Core ML guide, test plan, user guide, Mac guide, report, AI log, attribution, checklist
`scripts/`	verify-native.sh, generate-native-assets.mjs, build-docs-pdf.mjs

09

Limitations & roadmap

An honest status, and what comes next.

Current limitations

- Native iOS and visionOS code is not yet compiled — expect fixes on the first Xcode build
- No Core ML model is bundled, so identification runs in labelled demo mode
- Reminders fire at a fixed preferred hour; no weather or season awareness
- No iCloud sync between devices or between the iPhone and Vision Pro apps
- The spatial light estimate uses distance only
- Competitor details and market sizing need checking against current sources

Roadmap

1. Train and evaluate a Create ML model on a licensed dataset; publish real metrics
2. Core Data + CloudKit sync shared by iPhone and Vision Pro
3. Widgets and Live Activities for today's care
4. ARKit plane detection and light estimation in LeafLoom Spatial



LeafLoom

Grow with confidence.